Project 2

My project ended up being a very small kinda light ping pong ball launcher. The way I launched the balls is with two opposite opposed wheels spinning very fast. I turned it with a stepper motor. I was able to load the launcher with a servo that would move quickly up and down. The sensor was mounted on the moving part and would move back and forth 180 degrees and fire at anything in the area of 250 cm. The first challenge that I faced was how to move it. After testing we figured out that a single stepper motor would be the best bet. However after building my first plate it wouldn't turn no matter what. I ended up having to completely redo my first design just because of the stepper motor. Then came the firing mechanism. I built 4 different designs and only the last design worked properly. The first was a rubber band catapult. It didn't work due to how much tension I needed to launch the ball far enough. Then I tested this more with another rubber band launcher which was more like a bow on its side. This also didn't work because I didn't have motors strong enough to pull it back. I was able to get it working once but then it pulled the forward parts that the rubber band was attached to apart. I then tested using springs while this did work better and the motors were able to pull it back. I wasn't able to keep them compressed. I tried a lot of different methods, servos, different motors but nothing really worked. I then finally messed with the wheels. At first I used the gearbox but that didn't work at all. Not fast enough and stopped once the ball went through it. Then I had the idea to take them out of the gearbox and this worked. I didn't think it would work at all. Then after getting this all working i started testing everything individually and everything worked perfectly., then i put them together and that didn't work at all. My sensor would only scan once then get stuck in an infinite loop for the stepper motor. I tested a bunch of different methods of code but couldn't figure it out. I think it's a lack of knowledge on the coding side of things. I tried using the timer.h library and this I thought would work but I couldn't call on two things in it so it made the code not usable. Lastly, I had the idea that I would make a function and call it every step on the stepper motor function. This didn't work but it gave me an idea to call the code every few steps to then scan and that worked. Although not perfectly. It doesn't work perfectly in my book because I would have liked it to scan and run the stepper motor at the same time and launch at the same time. However I am limited by my knowledge of how the arduino program works overall.